

Ex.no:6b	DATA VISUALIZATION USING POWER BI (ADVANCED SALES ANALYTICS)

Aim:

To Load the data(Retail_sales) into PowerBI and transform the data and visualize it to get insights

Algorithm:**ALGORITHM 1: IMPORTING DATA INTO POWER BI****Step 1**

Open Power BI Desktop

Step 2

Select:

Home → Get Data

Step 3

Choose Excel or CSV

Step 4

Browse dataset

Step 5

Click Load

Step 6

Verify imported data

ALGORITHM 2: DATA TRANSFORMATION USING POWER QUERY**Step 1**

Select **Transform Data**

Step 2

Open Power Query Editor

Step 3

Remove duplicates

Step 4

Handle null values (replace blanks/nulls in columns like CustomerName and Discount)

Step 5

Rename columns & format text casing (e.g., Capitalize Region)

Step 6

Apply changes & Close

ALGORITHM 3: ADVANCED TRANSFORMATIONS & DATE PARSING

Step 1

Select OrderDate column and ensure data type is set to **Text**

Step 2

Go to **Add Column** → **Custom Column**

Step 3

Name column: ParsedDate

Step 4

Apply Power Query M Formula:

Code snippet

if Text.StartsWith([OrderDate], "2025-") then

 Date.FromText([OrderDate])

else if Text.Contains([OrderDate], "/") then

 Date.FromText([OrderDate], [Format = "dd/MM/yyyy", Culture = "en-GB"])

else

 Date.FromText([OrderDate], [Format = "MM-dd-yyyy", Culture = "en-US"])

Step 5

Handle negative quantities (Absolute Value or filter rows)

Step 6

Click **Close & Apply** to load the clean dataset into the model

ALGORITHM 4: CREATING CALCULATED COLUMN (REVENUE)

Step 1

Navigate to **Report View** or **Table View**

Step 2

Right-click dataset table and select **New Column**

Step 3

Enter Revenue formula:

$$\text{Revenue} = [\text{Quantity}] \times [\text{UnitPrice}] \times (1 - (\text{if } [\text{Discount}] = \text{null then } 0 \text{ else } [\text{Discount}]))$$

Step 4

Press **Enter** to validate and compute values

ALGORITHM 5: DASHBOARD VISUALIZATION & REPORTING

Step 1

Select **Line Chart** visualization and map ParsedDate (X-axis) and Revenue (Y-axis)

Step 2

Select **Bar / Column Chart** and map Region (X-axis) and Revenue / Quantity (Y-axis)

Step 3

Select **Donut Chart** to break down quantities by Product

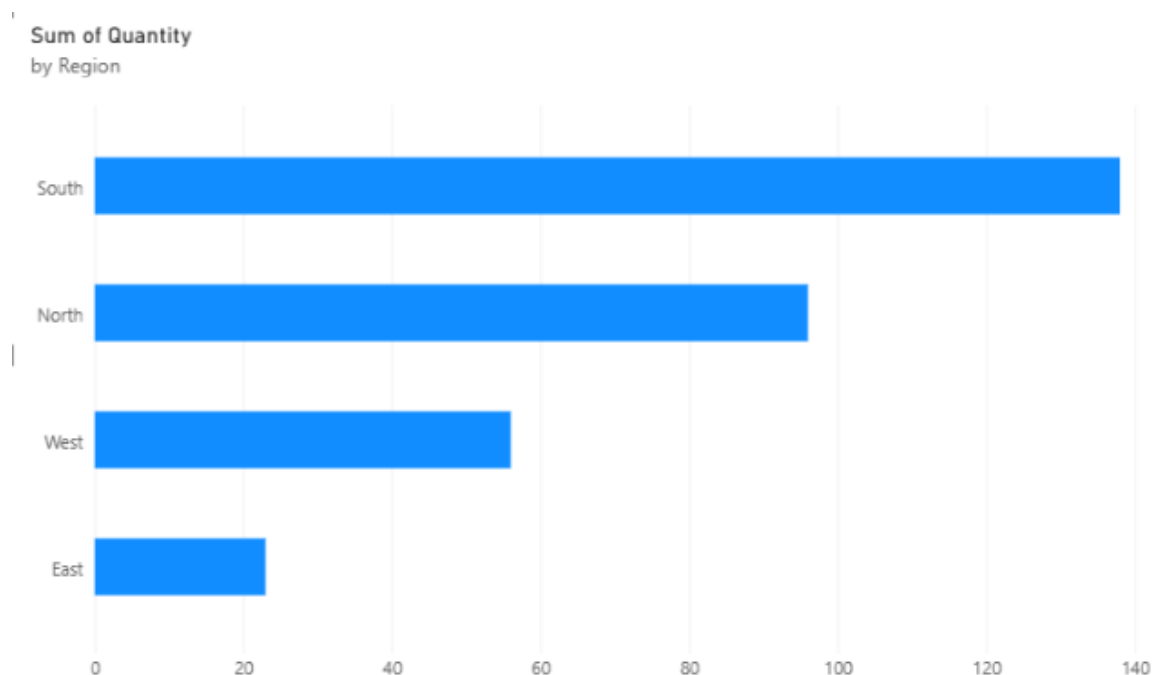
Step 4

Add **Slicer** visuals for Region and Date to enable interactive filtering

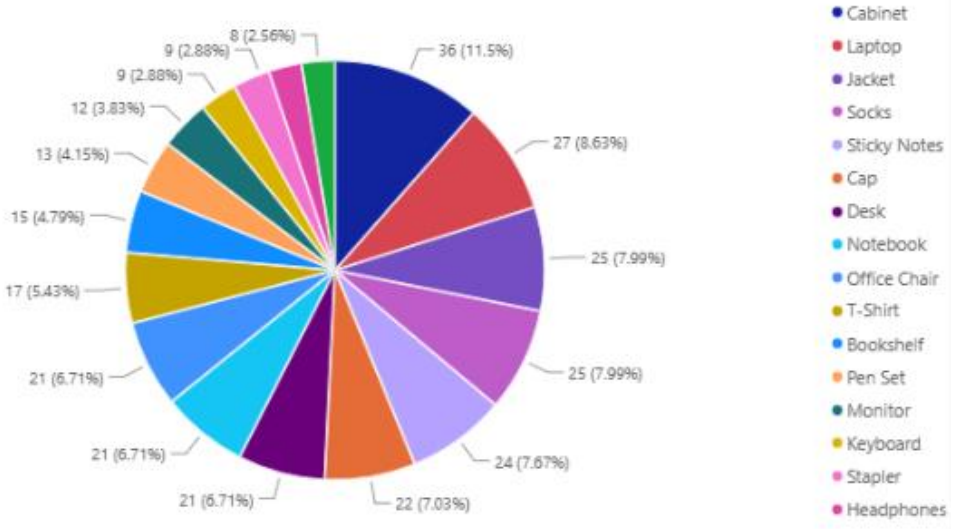
Step 5

Save the Power BI file (.pbix)

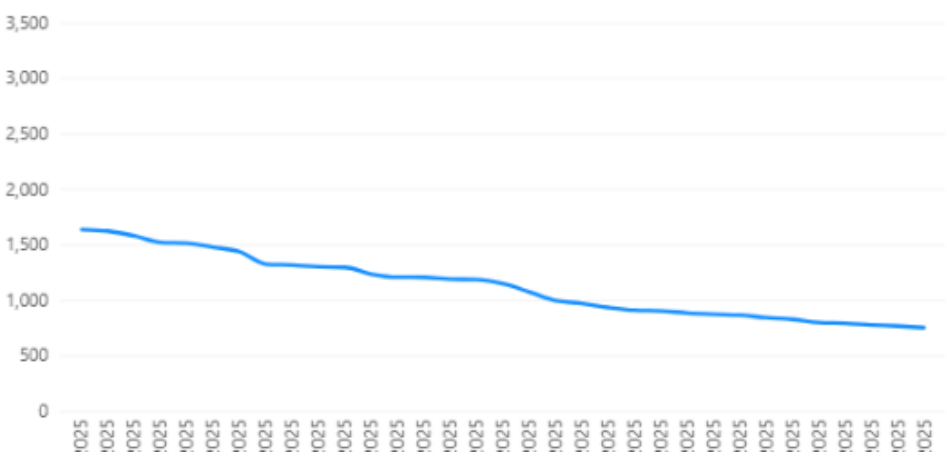
Output:



Sum of Quantity
by Product



Sum of Revenue
by ParsedDate

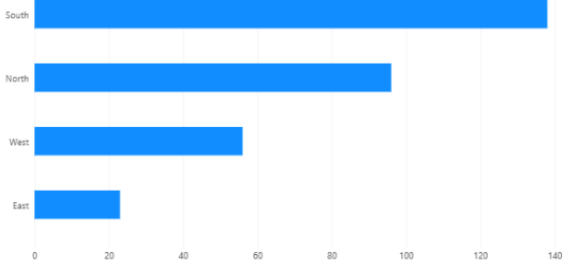


Sum of Revenue
74.31K

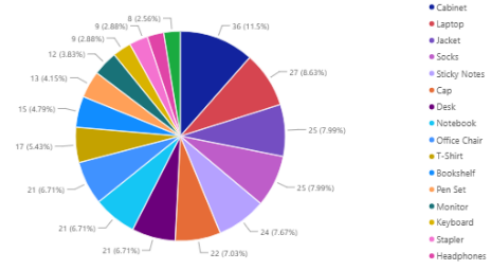
Region

- ☐ East
- ☐ North
- ☐ South
- ☐ West

Sum of Quantity
by Region



Sum of Quantity
by Product

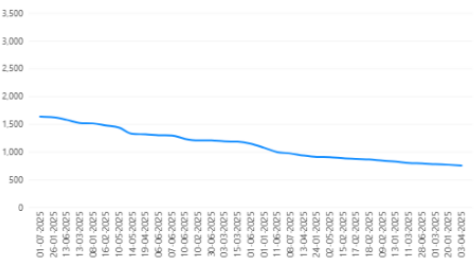


Sum of Revenue
74.31K

Region

- ☐ East
- ☐ North
- ☐ South
- ☐ West

Sum of Revenue
by ParsedDate



Result: